Every unconfined Steiner triple system carries an orientation

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Editorial note. The counted statement is the author's environment at source0.tex lines 445-446 and its proof environment opens at line 447; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial scope. One original result of the pinned author manuscript, counted once: the unconfined Steiner triple systems are oriented (unlabelled proposition at source0.tex lines 445-446).

Allowed standard prerequisites (editorial).

- Unconfined system. A Steiner triple system is unconfined when it carries an HF-ordering with no maximum.. It is the hypothesis of the inclusion.
- Orientation. An orientation of A selects one member of each block as its representative, with the closure condition on the resulting relation.. It is the structure that has to be produced.
- Inessential elements. An element belongs to at most one block of A when it is inessential, which makes it convenient to remove in the induction.. It is the case split of the first induction.

Proposition 0.1 $\mathcal{C}_{\mathrm{un}} \subseteq \mathcal{C}_{\mathrm{or}}$.

Proof. We check by induction on $|A|$ that if A is unconfined, then it has an orientation. We may assume $A \neq \emptyset$. Let $a \in A$ belong to at most one block of A. By induction hypothesis, $A \setminus \{a\}$ has an orientation c. If a belongs to the block e, we make a the apex of e. This gives an orientation of A extending c.

Now we prove by induction on $|A|$ that if A is unconfined and $X \subseteq A$, and $|X| \leq 2$, then $X \leq^* A$. It is enough to check that if $X \subseteq A$, then $\delta(X) \geq \min\{|X|, 2\}$. We may assume $A \neq \emptyset$. Choose $a \in A$ belonging to at most one block of A and let $A' = A \setminus \{a\}$. We can apply the induction hypothesis to A'. If $a \notin X$, then $X \subseteq A'$ and the induction hypothesis gives the result. If $a \in X$, let $X' = X \setminus \{a\}$. Then $\delta(X') = \delta(X)$ or $\delta(X) = \delta(X') + 1$. If $|X| \leq 2$, then $\delta(X) = |X|$ and clearly $\delta(X) \geq \min\{|X|, 2\}$. If $|X| > 2$, then $|X'| \geq 2$, and by induction hypothesis $\delta(X') \geq \min\{|X'|, 2\} = 2$, hence $\delta(X) \geq \delta(X') \geq 2 = \min\{|X|, 2\}$. □